Viral pathogenesis and immunity

VIRAL PATHOGENESIS AND IMMUNITY (sample notes)

Mechanisms of damage
- Direct cytopathic effect: lysis of infected cells or syncytia formation.
- Immune-mediated damage: cytotoxic T-cell killing of infected cells, or inflammatory responses.
- Oncogenesis: some viruses (for example, HPV) can drive uncontrolled cell growth.

Host defence
- Innate: interferons (type I) induce antiviral states in neighbouring cells; natural killer cells kill infected cells lacking MHC class I.
- Adaptive: neutralising antibodies block entry; CD8+ cytotoxic T cells eliminate infected cells; CD4+ helper cells coordinate the response.

Clinical link: vaccines work by priming neutralising antibodies and memory T cells without causing disease.